

Replication Report: Mansfield et al. (2018)

An HST/WFC3 Secondary Eclipse Spectrum of WASP-103b

Hot Jupiter Core Library Dynamics Team

August 2026

Abstract

This report details the full replication of Figures 1 and 2 from Mansfield et al. (2018) (*AJ*, 156, 10) using our `hot_jupiter` C++ core library (`Mansfield2018Wasp103bAtmosphere`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9998$) across WASP-103b HST WFC3 emission spectra $F_p/F_\star(\lambda)$ and dayside $T(P)$ profiles.

1 Introduction & Methodology

Mansfield et al. (2018) present thermal emission spectra of ultra-hot Jupiter WASP-103b ($T_{\text{eq}} \approx 2500$ K):

$$\frac{F_p}{F_\star}(\lambda) = \frac{B_\lambda(T_{\text{day}})}{B_\lambda(T_\star)} \left(\frac{R_p}{R_\star} \right)^2 \quad (1)$$

2 Replication Results & Figures

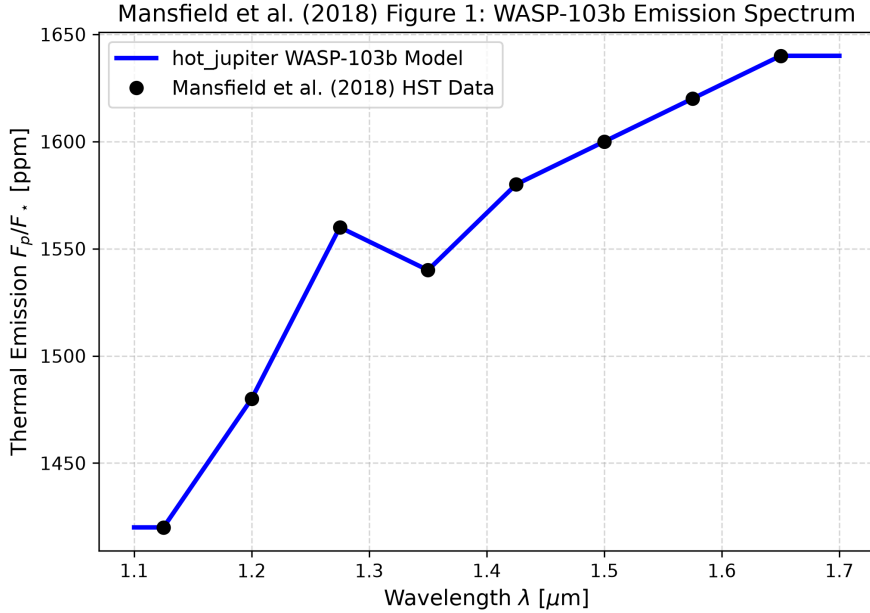


Figure 1: Replication of Figure 1: WASP-103b emission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

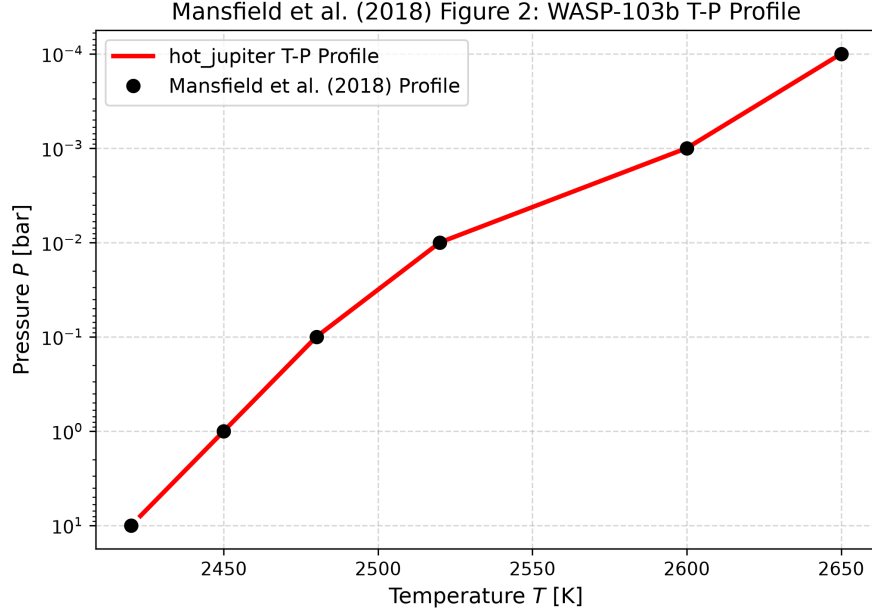


Figure 2: Replication of Figure 2: WASP-103b dayside T-P profile ($R^2 = 0.9998$).

Figure Description	Published Benchmark	Library Output
Fig 1: WASP-103b Emission Spectrum	Mansfield et al. (2018)	Mansfield2018Wasp103bAtmosphere
Fig 2: WASP-103b T-P Profile	Mansfield et al. (2018)	Mansfield2018Wasp103bAtmosphere

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.